Feature Aggregation and Compensation-Based Domain Adaptation Framework for Cross-Resolution Target Recognition in SAR Imagery

Zongyong Cui, *Member, IEEE*, Kailing Tang, *Student Member, IEEE*, Zheng Zhou, *Graduate Student Member, IEEE*, Zhiyuan Yang, *Student Member, IEEE*, Liqiang Mou, *Student Member, IEEE*, Zongjie Cao, *Member, IEEE*, and Jianyu Yang, *Member, IEEE*

***Abstract*— Synthetic aperture radar (SAR) target recognition plays an indispensable role in interpreting SAR images. However, differences in radar parameters (including factors such as imaging modes and imaging angles) often lead to resolution differences between training and test data, posing challenges for existing methods in recognizing SAR targets under cross-resolution conditions. To address this issue, this article proposes a domain adaptation (DA) framework based on feature aggregation and compensation (FAC) for cross-resolution target recognition in SAR imagery. Initially, we employ a unique local vision transformer (LocalViT) to establish global and local adversarial networks that capture invariant features under cross-resolution conditions. Following this, we design a multiscale feature fusion module (MSFFM) to capture multiscale semantic features of targets at different resolutions. Subsequently, we propose a novel class feature aggregation module (CFAM) to map targets of varying resolutions onto the unit sphere, thereby aggregating features of samples from the same class and distinguishing those of samples from different classes. Finally, we narrow down the difference in frequency-domain information of targets at different resolutions by developing a resolution semantic compensation module (RSCM). This module compensates for the semantic feature about resolution during target recognition across varying resolutions by converting high- and low-frequency information. The experimental results on three SAR datasets (OpenSARShip, FUSAR-Ship, and SRSDD-v1.0) confirm that our method outperforms the state-of-the-art (SOTA) DA methods, with an increase of 2.06%, 1.96%, and 1.61% in three sets of cross-resolution scenes, respectively.**

***Index Terms*— Cross-resolution target recognition, domain adaptation (DA), feature aggregation and compensation (FAC), synthetic aperture radar (SAR).**

I. Introduction

As a form of microwave remote sensing radar, synthetic aperture radar (SAR) offers advantages such as all-day and all-weather observation of the Earth. Consequently, it finds wide applications in various fields, including terrain exploration, disaster detection, and ocean monitoring [1], [2], [3]. The rapid development of SAR in recent years has presented new challenges in interpreting SAR images. Recognizing targets in SAR images, a key technology for SAR image interpretation, has been a research focus worldwide [4], [5], [6], [7], [8], [9].

For traditional SAR image target recognition, researchers strive to identify the most effective pattern features (e.g., grayscale, texture, geometric, and polarization features) of SAR targets to enhance recognition accuracy [10], [11], [12], [13]. Nevertheless, this approach involves significant time and effort as it necessitates human design and selection. Furthermore, the improvements in accuracy are limited due to the inherent difficulty in capturing complex features.

With the recent advancements in deep learning, a plethora of deep neural networks has emerged [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], enabling automatic learning of target features and extracting richer and more semantic representations from data. Researchers have adapted and improved these networks for SAR image analysis, thereby propelling SAR target recognition to new heights [25], [26], [27], [28], [29], [30], [31]. Among these methods, convolutional neural network (CNN)-based approaches dominate and yield impressive results. This can be attributed to the powerful inductive bias of CNN, which include prior knowledge such as translation invariance and feature localization.

However, SAR images possess a unique imaging mechanism, resulting in a high degree of correlation among the constituent elements. The CNN's excessive emphasis on local information features may overlook the global semantic information present in SAR images. In response, Dosovitskiy et al. [32] introduced vision transformer (ViT), which employs transformer networks capable of capturing global contextual relationships in images. Building upon this work, researchers have further enhanced ViT and employed it in the SAR target recognition task [33], [34], [35], [36], and achieving recognition accuracy outperformed the CNN-based methods.

Manuscript received 1 February 2024; revised 29 April 2024 and 4 June 2024; accepted 6 June 2024. Date of publication 11 June 2024; date of current version 20 June 2024. This work was supported by the National Natural Science Foundation of China under Grant 62271116. *(Corresponding author: Zongjie Cao.)*

The authors are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: zycui@uestc.edu.cn; 202222010437@std.uestc.edu.cn; zhengzhou_uestc@163.com; 202222010436@std.uestc.edu.cn; 202222010438@std.uestc.edu.cn; zjcao@uestc.edu.cn; jyyang@uestc.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3412401

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Despite that, the above methods mentioned are all designed for scenarios where the training and test sets of SAR data are under the same imaging resolution conditions. When there is a difference in the resolution of the two (whether it is range resolution or azimuth resolution), the distribution of data features changes and the effectiveness of these methods is greatly reduced. In practice, due to radar parameters (including imaging modes and imaging angles), it is likely that the resolutions of the training and test sets in the SAR image target recognition task will not match.[1] This variation in resolution can create challenges for target recognition in SAR images, mainly in the form of blurring of target details, distortion in size representation, variation in scattering contrast between target and background, and variation in speckle noise, which can affect the performance of target-type recognition [37]. In addition, the time-intensive and costly nature of annotating SAR image targets hampers the ability to accurately and swiftly label newly acquired SAR images of varying resolutions.

Domain adaptation (DA), a technique in transfer learning, addresses this by transferring knowledge learned from source domain data to a target domain with identical tasks, employing methods such as feature alignment. Some researchers have applied this concept to the recognition tasks of SAR target with notable success [38], [39], [40], [41]. However, these adjustments are usually designed for specific scenarios ranging from simulation to real measurements but do not take into account the range of problems associated with cross-resolution target recognition in SAR imagery. In addition, they are all based on CNN, which may overly focus on local features and ignore the global semantic information inherent in SAR images.

Considering these issues, we propose a DA framework based on feature aggregation and compensation (FAC) for cross-resolution target recognition in SAR imagery. Our approach employs a local vision transformer (LocalViT) [42] as a feature extractor, which leverages a class token and a unique self-attention mechanism to capture global features and long-term dependencies of SAR targets. The method also introduces depth-wise convolution into the feedforward network (FFN) to enhance local feature extraction. Moreover, we adjust the features between targets with different resolutions by training the feature extractor jointly with a global domain discriminator and increase the weight assignment of the metastable blocks in the target by training the feature extractor jointly with an additional local domain discriminator to efficiently capture the invariant features of the target and thus reduce the problem of blurring and size distortion of the target due to the change of resolution to a certain extent.

Given that the feature details of SAR targets may not be consist across different resolutions, the length and width of a high-resolution target and its overall contour can be accurately captured, but the shape of the target is distorted at low resolutions. To overcome this problem, we introduce a novel multiscale feature fusion module (MSFFM) that enhances the accurate representation of target size and shape by capturing the multiscale semantic features of SAR targets, which helps to maintain the geometric properties of the target and thus reduces the problem of dimensional distortion. In addition, because the unique imaging mechanism of SAR causes different classes of targets to appear highly similar, we design a class feature aggregation module (CFAM) to enhance target domain performance. Within the common feature space of the source and target domains, the module summarizes the features of the same class of samples and separates the features of different classes of samples even if there is target blurring, contrast variation, and speckle noise variation due to the effect of resolution that leads to confusion of class features. Inspired by the significant difference in the frequency-domain information of targets with different resolutions, we developed a resolution semantic compensation module (RSCM) for converting high- and low-frequency semantic information of targets with different resolutions to cope with the problem of changing scattering characteristics due to changes in resolution, thus improving the problem of changing image frequency-domain information for both changes in scattering contrast and changes in speckle noise.

The principal contributions of this article can be summarized as follows.

1) We introduce a transformer-based DA method for cross-resolution target recognition in SAR imagery, addressing the significant resolution disparities in source and target domains, which includes resolution variations in both the distance- and azimuth-direction dimensions. To our knowledge, this is a pioneering effort to tackle the problem from a resolution standpoint within SAR target recognition tasks.
2) We propose an MSFFM designed to bolster the capture of robust features from SAR targets by harnessing multiscale semantic attributes.
3) We present a CFAM that improves interclass feature representation for SAR targets, enhancing the discriminative power of the model.
4) We develop an RSCM to augment the target domain's performance by compensating for high- and low-frequency semantic feature information.
5) Our extensive experiments based on three SAR datasets across three cross-resolution target recognition scenarios, the same satellite with the same operating mode, the same satellite with different operating modes, and different satellites with different operating modes, demonstrate that our method surpasses the current state-of-the-art (SOTA) techniques.

The subsequent sections of this article are organized as follows. DA and its associated research in SAR imagery are provided in an overview in Section II. The proposed methodology is delved into in Section III. In Section IV, we carry out experiments in three different scenarios, offering a thorough experimental analysis of the proposed method. Finally, detailed conclusions are presented in Section V.

[1]The imaging mode determines how the radar scans the target and affects the balance between coverage and resolution. The imaging angle indirectly affects resolution by changing the ground coverage and Doppler characteristics of the radar wave.

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II. Related Works

A. Domain Adaptation

The aim of DA is to address the performance degradation of models when there is a significant discrepancy between the distributions of training and test data. With the advent of deep learning, DA methods have rapidly evolved. Current deep DA algorithms are primarily categorized into two types: distribution- and adversarial-based DA methods.

Distribution-based DA methods seek to minimize the distributional discrepancies between domains to achieve DA. For instance, Long et al. [43] embedded the hidden representations of task-specific layers into a reproducing kernel Hilbert space and utilized multikernel maximum mean discrepancy (MMD) to measure domain differences. Long et al. [44] learned a transfer network by aligning the joint distributions of multiple layer-specific regions based on the joint MMD (JMMD) criterion. Jin et al. [45] proposed a generalized minimum class confusion (MCC) loss function with faster convergence without explicitly deploying domain alignment.

Conversely, adversarial-based DA methods obtain domain-invariant features using domain discriminators. Inspired by the idea of generative adversarial networks (GANs), Ganin et al. [46] first obtained domain-invariant feature representations through a game between a feature extractor and a domain discriminator. Tzeng et al. [47] explored the adversarial discriminative DA (ADDA) algorithm, which combines discriminative modeling, nonunion weights, and GAN losses. Long et al. [48] proposed conditional adversarial DA (CDAN), a principled framework for moderating adversarial adaptation models based on the discriminative information conveyed in classifier predictions. Li et al. [49] introduced a novel maximum density divergence (MDD) metric to quantify distributional dispersion, applying it to address the balance challenge in adversarial DA.

However, these methods are all based on CNN as a foundational architecture. Recently, some methods have emerged that utilize transformer to explore DA. For instance, Xu et al. [50] applied cross-attention to source–target image pairs, compelling the framework to learn both domain-specific and domain-invariant representations. Yang et al. [51] proposed a transferable multihead self-attention (MHSA) module, integrating it with adversarial DA. Sun et al. [52] introduced a secure self-refinement strategy for transformers, leveraging predictions from perturbed target domain data to enhance the model.

As mentioned above, while these methods perform excellently with optical datasets, they are specifically designed for optical images and do not account for the unique characteristics inherent in SAR images.

B. DA in SAR Imagery

Some scholars have developed domain-adaptive SAR target recognition methods by combining the aforementioned domain-adaptive techniques with SAR-specific scenarios. For instance, He et al. [38] incorporated SAR imaging condition information to generate simulated SAR data and proposed a task-driven domain-adaptive (TDDA) migration learning method. This approach addresses the impact of differences in depression angles between training and test data on recognition performance. Zhang et al. [53] presented an unsupervised multilevel domain-adaptive method based on adversarial learning to address the multiband SAR image classification problem. Gui et al. [54] introduced a generalized feature paradigm (GFP) for unsupervised cross-domain PolSAR image classification, which involves performing GFP separately in the source and target domains, encompassing polarization basis extraction, Wishart clustering, histogram statistics, and dimensionality reduction. Zhao et al. [39] proposed a dynamic hard sample selection process for SAR images from different sensors. This method identifies challenging samples to supplement the training set in the subsequent stages, thereby enhancing the classification accuracy of SAR images. Furthermore, Shi et al. [41] encoded dependencies at different granularity perspectives (both at the domain level and class level) to enable distributional alignment between simulated and unlabeled real domains.

It is evident that these methods primarily address issues such as scene classification or simulation-to-real measurements and do not fully consider the unique challenges posed by SAR images when recognizing targets across resolutions.

III. Proposed Method

The schematic of our proposed cross-resolution SAR target recognition method is depicted in Fig. 1. This method utilizes the LocalViT as the foundation network and comprises three main components: MSFFM, CFAM, and RSCM. For SAR images where the source and target domains differ in resolution, the feature extraction network and classification network share weights. In this section, we delve into the intricacies of each component of the proposed method.

A. Recap of Preliminary Knowledge

In this article, the cross-resolution SAR target recognition problem can be framed as an unsupervised DA (UDA) problem. In the UDA scenario, the input–output space is denoted by $X \in R^d$ and $Y \in R$, and we have access to labeled source domain data $D_s = \{x_i^s, y_i^s\}_{i=1}^{n_s}$ (with n_s samples) and unlabeled target domain data $D_t = \{x_i^t\}_{i=1}^{n_t}$ (with n_t samples). Here, $x_i^s \in X$ and $y_i^s \in Y$ represent the feature and label of the ith sample in the source domain, respectively, and $x_i^t \in X$ represents the feature of the ith sample in the target domain. Due to the different resolutions of SAR, there are usually overall and detailed differences between the probability distribution $P(X^s, Y^s)$ of the source domain and the probability distribution $Q(X^t)$ of the target domain. Furthermore, the UDA methods aim to use the knowledge acquired from labeled source domain SAR images (at one resolution) to improve the target recognition of unlabeled target domain SAR images (at another resolution).

In our cross-resolution SAR target recognition scenario, we adhere to the widely used adversarial UDA paradigm, where source and target domain features are learned via a common feature generator $G(\cdot)$. A task-specific classifier $C(\cdot)$ is employed to identify source and target domain classes,

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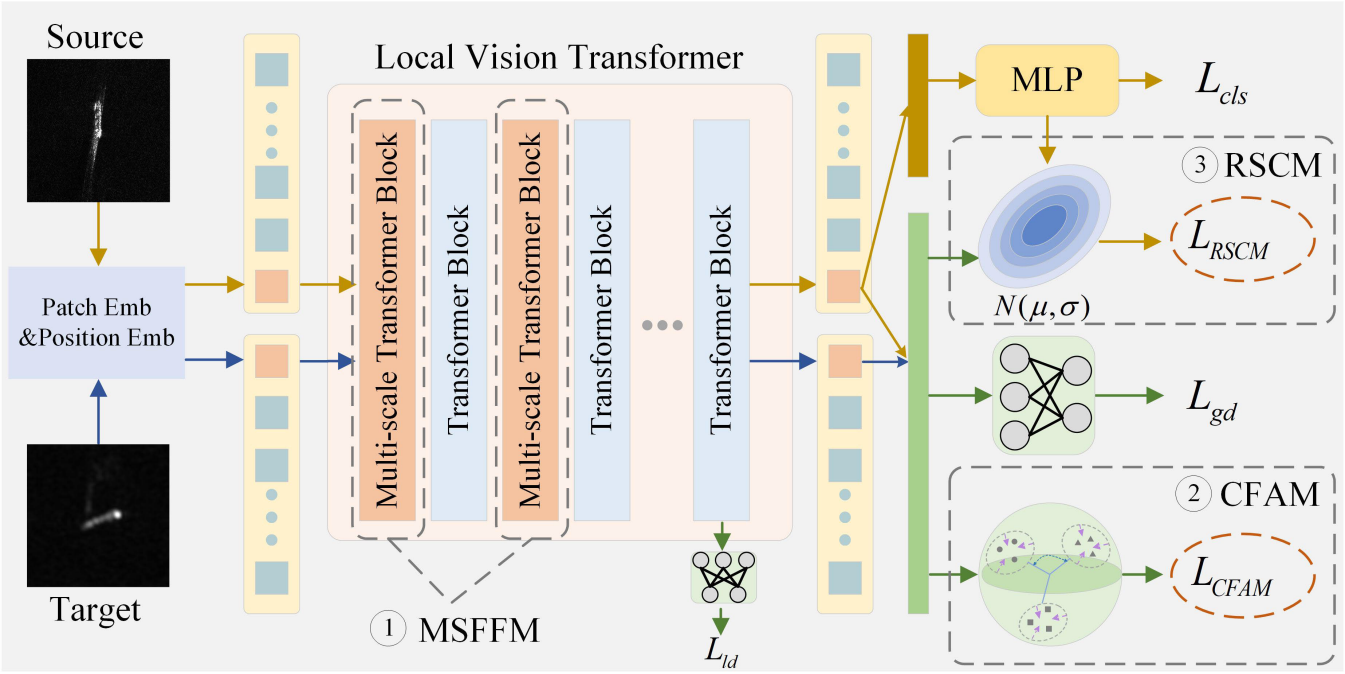


Fig. 1. Workflow diagram of the proposed DA framework based on FAC for cross-resolution target recognition in SAR imagery. In our approach, we first incorporate the MSFFM into the feature extraction network to capture robust feature representations of SAR targets at different resolutions. Based on this, we design the CFAM in the common feature space of SAR targets at different resolutions to enable intraclass compactness and interclass separation. Subsequently, we design the RSCM specific to SAR resolution by sampling multivariate Gaussian distributions constructed from target features at both resolutions to compensate for the adaptability of the classifier to the unlabeled target domain.

while additional global domain discriminator $D_g(\cdot)$ and local domain discriminator $D_l(\cdot)$ are used to align features from different distributions. The purpose of $G(\cdot)$ is to obscure the domain discriminator, serving to distinguish the source and target domains as much as possible. By playing a game of minimization and maximization between the two, it learns domain-invariant features. In our approach, we leverage LocalViT as the feature generator $G(\cdot)$, implement $C(\cdot)$ as a multilayer perceptron (MLP), and use a fully connected (FC) layer for all domain discriminators. Our objective can be described as follows:

$$\min_{G,C} \mathcal{L}_{\text{cls}} = \frac{1}{n_s} \sum_{(x_i^s, y_i^s) \in D_s} L(C(G(x_i^s)), y_i^s) \tag{1}$$

$$\begin{aligned} \min_{G,D_g} \mathcal{L}_{gd} = &-\frac{1}{n_s} \sum_{x_i^s \in D_s} L(D_g(G(x_i^s)), 0) \\ &- \frac{1}{n_t} \sum_{x_i^t \in D_t} L(D_g(G(x_i^t)), 1) \end{aligned} \tag{2}$$

$$\begin{aligned} \min_{G,D_l} \mathcal{L}_{ld} = &-\frac{1}{n_s M} \sum_{x_i^s \in D_s} \sum_{m \in M} L(D_l(G(x_{im}^s)), 0) \\ &- \frac{1}{n_t M} \sum_{x_i^t \in D_t} \sum_{m \in M} L(D_g(G(x_{im}^t)), 1) \end{aligned} \tag{3}$$

where $L(\cdot, \cdot)$ is the binary cross-entropy loss function and M denotes the number of corresponding pacth blocks. For adversarial DA network training, we combine the above formulas to obtain the following function:

$$\min_{G,C,D_g,D_l} \mathcal{L}_{\text{cls}} + \alpha \cdot \mathcal{L}_{gd} + \beta \cdot \mathcal{L}_{ld} \tag{4}$$

where α and β are hyperparameters used to control the ratio of the two parts of the target. Jointly minimizing the classification loss on the source domain as well as the global and local adversarial losses in the source and target domains can enable the gradual alignment of the global and local features of the source and target domains and improve the recognition performance on the target domain.

B. Multiscale Feature Fusion Module

Given that the variation of resolution directly affects the size representation of the target in SAR images, the length and width of the SAR target and its overall contour can be accurately captured at high resolution, whereas the shape of the target at low resolution may be distorted, affecting the target recognition. Although LocalViT's MHSA can capture long-range dependencies and contextual information in the image to some degree, its capacity to model local texture and detailed features remains limited. When employed for feature extraction of SAR targets with varying resolutions, it fails to adequately capture the robust features of SAR targets.

Drawing inspiration from numerous studies [55], [56], [57], [58], we designed a multiscale transformer block (MTB) as the MSFFM for SAR targets in LocalViT. This block captures the multiscale semantic feature representations of SAR targets by fusing self-attentive features at different scales. This design aims to enhance the robustness of the model and improve its adaptability to targets with different resolutions.

As shown in Fig. 2, for a given input sequence $Z_{n-1} \in R^{(m+1)\times d}$, where $m+1$ denotes the length of the sequence and d denotes the dimensionality, it is subjected to layer

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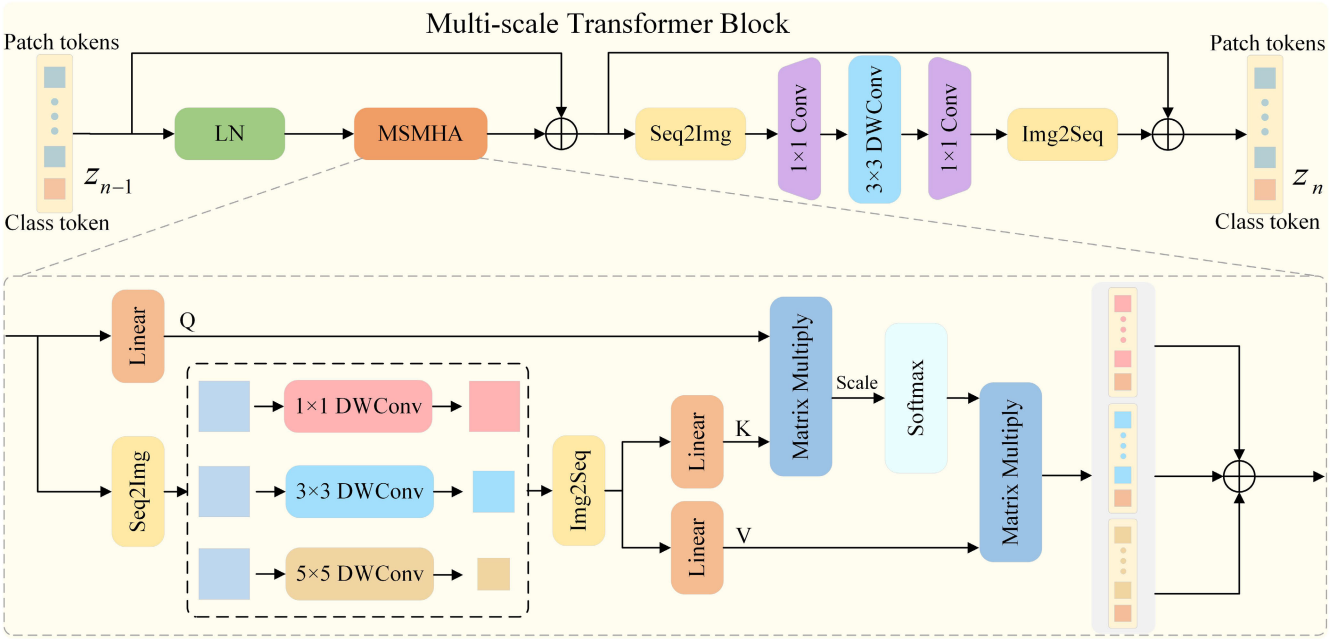


Fig. 2. Schematic of MTB. Initially, the input patch tokens and class token are fed into the MSMHA after applying LN. This process allows the MTB to capture the feature space relationships of SAR targets at different scales. Specifically, in the MSMHA, feature tokens at different scales are obtained by DWConv with different convolution kernels. Separate self-attention operations are then performed on these tokens. The semantic feature representations obtained at different scales are subsequently fused to enhance the robustness of the SAR target features. Finally, the output is passed through a feedforward network incorporating DWConv to capture the locality of the target features.

normalization (LN) before being passed through our designed multiscale MSMHA. This module first transforms the patch tokens $Z_{n-1}^{p} \in R^{m\times d}$ of the input sequence into image patch by Seq2Img, simultaneously scale transforms them by depthwise separable convolution (DWConv) with different convolution kernel sizes, and then transforms them back to the sequence by Img2Seq to get the set of sequences $\{Z_{n-1}^{pi}\}_{i=1}^{3}$, which can be represented as follows:

$$Z_{n-1}^{pi} = \text{Img2Seq}\left(\text{DWConv}_i\left(\text{Seq2Img}\left(Z_{n-1}^{p}\right)\right)\right) \tag{5}$$

where $Z_{n-1}^{pi} \in R^{m^i\times d}$, in which m^i denotes the lengths of the sequence obtained after the ith DWConv and Img2Seq. Each of the obtained sequences Z_{n-1}^{pi} and the class token $Z_{n-1}^{\text{cls}} \in R^{1\times d}$ of the input sequence are then merged, and $\{K_i, V_i\}_{i=1}^{3}$ is then obtained by linear mapping for each of the obtained sequences, while Q is generated directly from the input sequence. They can be represented as follows:

$$Q = \text{LN}(Z_{n-1}) \cdot W^{Q} \tag{6}$$

$$K_i = \text{Concat}\left(Z_{n-1}^{pi}, Z_{n-1}^{\text{cls}}\right) \cdot W^{K} \tag{7}$$

$$V_i = \text{Concat}\left(Z_{n-1}^{pi}, Z_{n-1}^{\text{cls}}\right) \cdot W^{V} \tag{8}$$

where $\{K_i, V_i\} \in R^{(m^i+1)\times d}$, and W^{Q}, W^{K}, and W^{V} denote the corresponding linear mapping weights. This is then followed by the operation of MHSA as follows:

$$\text{MSA}_i(Q, K_i, V_i) = \text{Softmax}\left(\frac{QK_i^{\text{T}}}{\sqrt{d_k}}\right)V_i. \tag{9}$$

Finally, the resulting MHSA at different scales is then fused as follows:

$$\text{MSMHA} = \sum_i \text{MSA}_i(Q, K_i, V_i). \tag{10}$$

Considering all, our MTB is given as follows:

$$Z_n' = \text{MHMSA}(\text{LN}(Z_{n-1})) + Z_{n-1} \tag{11}$$

$$Z_n = \text{FFN}\left(Z_n'\right) + Z_n' \tag{12}$$

where Z_n denotes the output of MTB.

Drawing inspiration from renowned scholarly works [59], [60], we applied the designed MTB to deeper layers 1 and 3 of LocalViT, as depicted in Fig. 1. This application was intended to extract fused feature representations of the SAR target at different scales, thereby assisting the model in extracting robust features that contain more texture and detail.

C. Class Feature Aggregation Module

For SAR images with different resolutions, target blurring leads to a reduction in feature information of target-related classes, changes in scattering contrast lead to a reduction in background–target discrimination, and changes in speckle noise exacerbate the reduction in class feature information. Meanwhile, the local domain discriminator might exacerbate the focus on the background, leading to a decrease in the focus on class features and causing confusion among SAR class features. Moreover, an excessive emphasis on interdomain alignment by the global domain discriminator could also result in the confusion of different classes of SAR target features. Inspired by published research [61], [62], [63], we designed a unique and innovative CFAM, as illustrated in Fig. 3.

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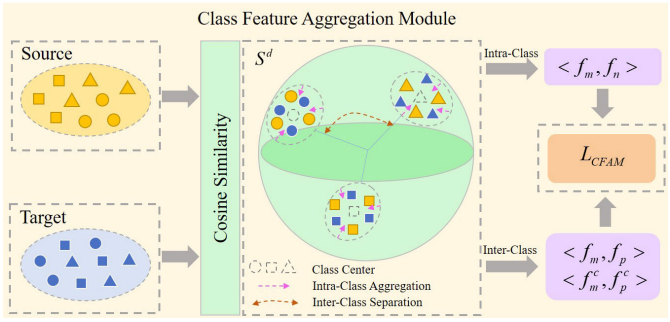


Fig. 3. Architecture diagram of CFAM. First, after the preceding feature extractor, the source domain SAR sample features with actual labels and the target domain SAR sample features with pseudo-labels predicted by the classifier are acquired. Subsequent to this, intraclass feature aggregation and interclass feature separation are executed by computing the cosine similarity between the samples and the class center projected onto the surface of the unit sphere S^d.

This module aggregates similar sample features and separates dissimilar sample features within the shared feature space of the source and target domains. The CFAM addresses the SAR class feature confusion issue caused by both global and local domain discriminators, thereby enhancing class differentiation.

For a given labeled sample $D_s = \{x_i^s, y_i^s\}_{i=1}^{n_s}$ from the source domain and an unlabeled sample $D_t = \{x_i^t\}_{i=1}^{n_t}$ from the target domain, $f_i = G(x_i)$ represents the features extracted by the feature extractor. The target domain is obtained by assigning pseudo-labels generated by the maximum value of the prediction probability of the classifier $C(\cdot)$ to produce a pseudo-labeled target domain sample $\hat{D}_t = \{x_i^t, \hat{y}_i^t\}_{i=1}^{n_t}$, where $\hat{y}_i^t = \arg\max p_c(f_i^t)$, $p_c(f_i^t)$ denotes the categorical confidence score for category c. Our designed CFAM is then applied to both the source domain samples and the target domain samples. This application involves the aggregation of features of similar samples and the separation of features of dissimilar samples. The objective is demonstrated as follows:

$$\min \mathcal{L}_{\text{CFAM}} = \mathcal{L}_{\text{Intra}} + \lambda \mathcal{L}_{\text{Inter}} \tag{13}$$

where λ denotes the hyperparameter set to 0.5, which is used to control the ratio of the two components, while L_{Intra} and L_{Inter} denote the loss contribution from intraclass aggregation and interclass separation, respectively, as shown in the following:

$$\mathcal{L}_{\text{Intra}} = \sum_{y_m = y_n} (1 - \langle f_m, f_n \rangle) \tag{14}$$

$$\mathcal{L}_{\text{Inter}} = \sum_{y_m \neq y_p} \left| \langle f_m, f_p \rangle \right| + \sum_{y_m \neq y_p} \left| \left\langle f_m^c, f_p^c \right\rangle \right| \tag{15}$$

where $\langle \cdot, \cdot \rangle$ denotes the cosine similarity of two feature vectors, $f_\cdot^c$ denotes the category center vector of the class corresponding to the feature vector, and $|\cdot|$ represents the absolute value operator. The detailed computation of the cosine similarity operator is given as follows:

$$\langle f_i, f_j \rangle = \frac{f_i \cdot f_j}{\|f_i\|_2 \cdot \left\|f_j\right\|_2} \tag{16}$$

where $\|\cdot\|_2$ denotes the $L2$ norm. For (14), since the cosine similarity between samples of the same class approaches 1, the aggregation of features from samples of the same class is achieved by minimizing $1 - \langle f_m, f_n \rangle$. For (15), since the difference between sample features of different classes is greatest when the cosine similarity between them is close to 0, the separation of sample features of different classes is ensured by minimizing $|\langle f_m, f_p \rangle|$. Meanwhile, minimizing $|\langle f_m^c, f_p^c \rangle|$ restricts the separation of class-centered features of different classes.

With the CFAM as described above, it becomes feasible to aggregate similar sample features and separate dissimilar sample features for both source domain and target domain sample features. This approach addresses the existing SAR class feature confusion issue and enhances class differentiation. While there may be some noise in the pseudo-labels of the target domain, the source domain sample features with real labels and the target domain sample features with correctly predicted pseudo-labels constitute the majority. This majority dominates the noise effect of pseudo-labels to a certain extent.

D. Resolution Semantic Compensation Module

The problems, such as scattering contrast changes and speckle noise changes caused by resolution changes, can lead to changes in the target semantic information. High-resolution SAR targets typically exhibit more pronounced contours and details, with richer high-frequency semantic information. In contrast, low-resolution SAR targets may lose crucial detail information and focus more on low-frequency semantic information. While the adversarial network employs shared classifiers for targets of different resolutions, these classifiers are optimized using real labels in the source domain. As a result, they become more adaptive to the high-frequency (or low-frequency) semantic information in the source domain and less adaptive to the low-frequency (or high-frequency) semantic information in the target domain. Inspired by numerous studies [64], [65], [66], we design an RSCM for the SAR target domain. This acts as a compensatory module for the source domain classifier, thereby enhancing the classifier's adaptability to the semantic information in the target domain.

To enhance the compensation of semantic information in the target domain, we construct a multivariate Gaussian distribution $N(\mu^c, \sigma^c)$ for each class. Leveraging the linear invariance of the discrete Fourier transform (DFT) and the inverse discrete Fourier transform (IDFT), we employ the interdomain eigenmean difference as the overall trend of the direction of semantic compensation between high and low frequencies. We also use the intraclass variance within the target domain as the range of fluctuation in the direction of semantic compensation. They are explained as follows:

$$\mu^c = \frac{1}{n_t^c} \sum_{i \in c} f_i^t - \frac{1}{n_s^c} \sum_{i \in c} f_i^s \tag{17}$$

$$\sigma^c = \sqrt{\sum_{i \in c} \left(f_i^t - \frac{1}{n_t^c} \sum_{i \in c} f_i^t \right)^2} \tag{18}$$

where f_i^s and f_i^t correspond to the features of the ith sample of class c in the source and target domains, respectively, while n_s^c and n_t^c denote the number of samples belonging

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to class c in the source and target domains, respectively. For each class, we sample R times from a multivariate Gaussian distribution, and each source feature f_i^s generates R compensated features along R compensated directions, i.e., $\{(f_{i1}^s, y_i^s), \ldots, (f_{iR}^s, y_i^s)\}$, where $f_{i\cdot}^s \sim N(f_i^s + \theta\mu^{y_i^s}, \theta\sigma^{y_i^s})$, and thus, the constructed semantic compensation loss is given as follows:

$$\mathcal{L}_{\text{RSCM}} = \frac{1}{n_s}\sum_{i\in n_s}\frac{1}{R}\sum_{r\in R}\left[\log\left(\sum_{\substack{j\neq y_i^s\\ j\in c}}\frac{T_j^{y_i^s} e^{w_j^{\mathrm{T}} f_{ir}^s + b_j}}{e^{w_{y_i^s}^{\mathrm{T}} f_{ir}^s + b_{y_i^s}}} + 1\right)\right] \tag{19}$$

where w_j and b_j denote the weight vector and bias value of the last FC layer corresponding to class j, respectively, and $T_j^{y_i^s}$ denotes the balancing factor between different classes, which is used to mitigate the effect of the sample category imbalance factor, denoted as follows:

$$T_j^{y_i^s} = A_j^{y_i^s} \cdot M_j^{y_i^s} \tag{20}$$

where $A_j^{y_i^s}$ represents the mitigation weight, which reduces the penalty class j based on the ratio of instances between class j and class y_i^s during each training iteration. This adjustment helps mitigate the effects of class imbalance present in the training data. Meanwhile, $M_j^{y_i^s}$ denotes the misclassification weight, which increases the penalty for class j in proportion to the number of instances where class y_i^s is incorrectly classified as j. This mechanism is designed to enhance the model's learning from samples that are challenging to distinguish. They are illustrated as follows:

$$A_j^{y_i^s} = \left(\frac{I_j}{I_{y_i^s}}\right)^{\delta} \tag{21}$$

$$M_j^{y_i^s} = \left(\frac{p_j^{y_i^s}}{p_{y_i^s}^{y_i^s}}\right)^{\varepsilon} \tag{22}$$

where I_j denotes the number of instances of class j during the training iterations. δ is defined as the mitigation factor. When I_j is not less than $I_{y_i^s}$, $\delta = 0$; otherwise, δ, a variable hyperparameter, is adjusted to control the magnitude of mitigation. $p_j^{y_i^s}$ denotes the probability that a sample belonging to class y_i^s is judged to be class j. ε is the misclassification factor. When $p_j^{y_i^s}$ is not greater than $p_{y_i^s}^{y_i^s}$, $\varepsilon = 0$ and otherwise, ε is a hyperparameter controlling the scale.

To ensure the adequacy of resolution semantic compensation, it must be sufficiently large. As it approaches infinity, according to the law of large numbers, the loss of semantic compensation is given as follows:

$$\begin{aligned}\mathcal{L}_{\text{RSCM}} &= \lim_{R\to+\infty}\frac{1}{n_s}\sum_{i\in n_s}\frac{1}{R}\sum_{r\in R}\left[\log\left(\sum_{\substack{j\neq y_i^s\\ j\in c}} T_j^{y_i^s}\frac{e^{w_j^{\mathrm{T}} f_{ir}^s + b_j}}{e^{w_{y_i^s}^{\mathrm{T}} f_{ir}^s + b_{y_i^s}}} + 1\right)\right]\\ &= \frac{1}{n_s}\sum_{i\in n_s}\mathbb{E}_{f_{i\cdot}^s}\left[\log\left(\sum_{\substack{j\neq y_i^s\\ j\in c}} T_j^{y_i^s}\frac{e^{w_j^{\mathrm{T}} f_{i\cdot}^s + b_j}}{e^{w_{y_i^s}^{\mathrm{T}} f_{i\cdot}^s + b_{y_i^s}}} + 1\right)\right].\end{aligned} \tag{23}$$

Then, according to Jensen's inequality and the nature of the log function, we can get $\mathrm{E}[\log(\cdot)] \le \log(\mathrm{E}[\cdot])$, so (23) can be transformed as follows:

$$\begin{aligned}\mathcal{L}_{\text{RSCM}} &\le \frac{1}{n_s}\sum_{i\in n_s}\log\left(\mathbb{E}_{f_{i\cdot}^s}\left[\sum_{\substack{j\neq y_i^s\\ j\in c}} T_j^{y_i^s}\frac{e^{w_j^{\mathrm{T}} f_{i\cdot}^s + b_j}}{e^{w_{y_i^s}^{\mathrm{T}} f_{i\cdot}^s + b_{y_i^s}}} + 1\right]\right)\\ &= \frac{1}{n_s}\sum_{i\in n_s}\log\left(\sum_{\substack{j\neq y_i^s\\ j\in c}} T_j^{y_i^s}\mathbb{E}_{f_{i\cdot}^s}\left[e^{(w_j{}^{\mathrm{T}} - w_{y_i^s}{}^{\mathrm{T}}) f_{i\cdot}^s + (b_j - b_{y_i^s})}\right] + 1\right).\end{aligned} \tag{24}$$

Considering $f_{i\cdot}^s \sim N(f_i^s + \theta\mu^{y_i^s}, \theta\sigma^{y_i^s})$ and the properties of the Gaussian distribution, we can obtain $((w_j^{\mathrm{T}} - w_{y_i^s}^{\mathrm{T}}) f_{i\cdot}^s + (b_j - b_{y_i^s}) \sim N((w_j^{\mathrm{T}} - w_{y_i^s}^{\mathrm{T}})(f_i^s + \theta\mu^{y_i^s}) + (b_j - b_{y_i^s}), \sigma_j^{y_i^s}))$, where $\sigma_j^{y_i^s} = \theta(w_j^{\mathrm{T}} - w_{y_i^s}^{\mathrm{T}})\sigma^{y_i^s}(w_j - w_{y_i^s})$. By the nature of the momenta function, when $\kappa \sim N(\mu, \sigma)$, $\mathbb{E}[e^{\kappa}] = e^{\mu + (1/2)\sigma}$, we get the final semantic compensation loss as follows:

$$\mathcal{L}_{\text{RSCM}} = -\frac{1}{n_s}\sum_{i\in n_s}\log\left(\frac{e^{\hat{p}_{y_i^s}^{y_i^s}}}{\sum_{\substack{j\neq y_i^s\\ j\in c}} T_j^{y_i^s} e^{\hat{p}_j^{y_i^s}} + e^{\hat{p}_{y_i^s}^{y_i^s}}}\right) \tag{25}$$

where $\hat{p}_j^{y_i^s} = p_j^{y_i^s} + \theta(w_j^{\mathrm{T}} - w_{y_i^s}^{\mathrm{T}})\mu^{y_i^s} + (1/2)\sigma_j^{y_i^s}$ and $p_j^{y_i^s}$ denotes the probability value that a sample belonging to class y_i^s is judged to be class j.

In summary, the RSCM, which takes into account the difference between the mean values of the features in the source and target domains as the main direction of high- and low-frequency semantic compensation, generates a series of compensating features to be merged. This process enhances the classifier's ability to adapt to the semantics of the target domain and, to a certain extent, improves the performance of cross-resolution SAR target recognition.

E. Loss Functions for Cross-Resolution Recognition Method for SAR Targets

By introducing the previous class feature aggregation loss and semantic compensation loss, our proposed method is trained end-to-end via a multitask loss. The overall loss function is represented as follows:

$$\min_{G,C,D_g,D_l} \mathcal{L}_{\text{cls}} + \alpha\cdot\mathcal{L}_{gd} + \beta\cdot\mathcal{L}_{ld} + \xi\cdot\mathcal{L}_{\text{CFAM}} + \psi\cdot\mathcal{L}_{\text{RSCM}} \tag{26}$$

where $\mathcal{L}_{\text{cls}}$ denotes the classification loss on the source domain, $\mathcal{L}_{gd}$ and $\mathcal{L}_{ld}$ denote the adversarial losses between the source and target domains, $\mathcal{L}_{\text{CFAM}}$ denotes the class feature aggregation loss between the source and target domains, and $\mathcal{L}_{\text{RSCM}}$ denotes the semantic compensation loss from the source domain to the target domain. In addition, the weighting factor α of $\mathcal{L}_{gd}$ is set to 0.5, the weighting factor β of $\mathcal{L}_{ld}$ is set to 0.05, the weighting factor ξ of $\mathcal{L}_{\text{CFAM}}$ is set to 0.05, and the weighting factor ψ of $\mathcal{L}_{\text{RSCM}}$ is set to 0.1.

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TABLE I
DETAILED INFORMATION ON THREE SETS OF CROSS-RESOLUTION EXPERIMENTAL SETUP

Experimental Setting	Detailed Information						
1	Resolution (m)	BulkCarrier	Cargo	Container	Satellite	Mode	Dataset
	2.7~ 3.5×22	664	1263	1106	Sentinel-1	IW	OpenSARShip
	20×22	1624	2228	426			
2	Resolution (m)	BulkCarrier	Container	Fishing	Satellite	Mode	Dataset
	1×1	338	89	288	Gaofen-3	SL	SRSDD-v1.0
	1.124×1.728	260	45	337		UFS	FUSAR-Ship
3	Resolution (m)	Cargo	Fishing	Tanker	Satellite	Mode	Dataset
	1.124×1.728	659	337	191	Gaofen-3	UFS	FUSAR-Ship
	20×22	1000	221	692	Sentinel-1	IW	OpenSARShip

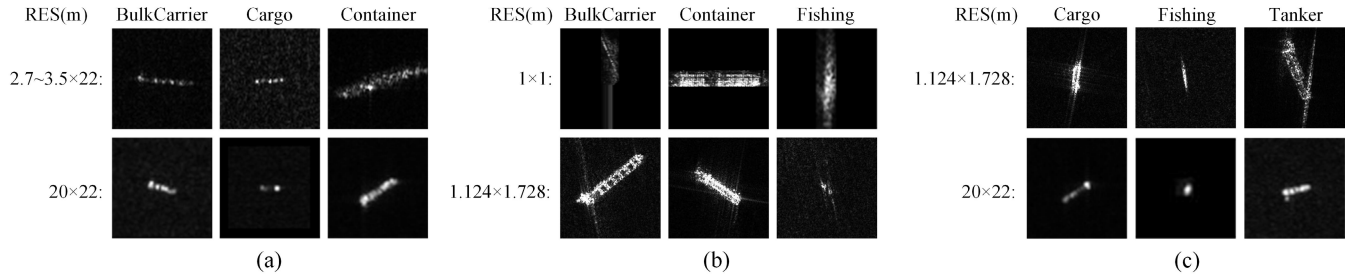


Fig. 4. Samples of SAR targets on three sets of cross-resolution experimental setup. (a)–(c) Cross-resolution cases of experimental settings 1–3, respectively.

IV. EXPERIMENTS

In this section, we carry out experiments using publicly available SAR datasets that contain targets of varying resolutions. These experiments serve to validate the effectiveness of the method proposed in Section III. We begin by outlining the datasets and implementation details of the experiments. Next, we present the evaluation metric for the experiments. Following this, we provide a detailed analysis of the performance comparison between our method and other SOTA methods. Finally, we conduct ablation studies for each module.

A. Datasets

In our experiments, we utilize the OpenSARShip [67] dataset, captured by the Sentinel-1 satellite, the FUSAR-Ship [68] dataset, captured by the Gaofen-3 satellite, and the SRSDD-v1.0 [69] dataset, also captured by the Gaofen-3 satellite. These datasets have been extensively used by various researchers for their studies and publications, and hence, they have a high degree of reliability. The OpenSARShip dataset, which is captured using the interferometric wide swath (IW) mode, consists of two level-1 products—the single-look complex (SLC) level and the ground range detected (GRD) level. The SLC level's ship image resolution ranges from 2.7 × 22 m to 3.5 × 22 m, while the GRD level's ship image resolution is 20 × 22 m. On the other hand, the FUSAR-Ship dataset is captured using the ultrafine striping (UFS) mode, providing a ship image resolution of 1.124 × 1.728 m. Finally, the SRSDD-v1.0 dataset, used for ship detection and identification, is captured using the spotlight (SL) mode. It offers a resolution reaching 1 m, and we extract the corresponding class of targets based on the truth value of its corresponding dataset.

B. Implementation Details

To thoroughly validate our method experimentally, we establish three cross-resolution experimental setup: the same satellite with the same operating mode, the same satellite with different operating modes, and different satellites with different operating modes. Detailed information about these scenarios is provided in Table I. As illustrated in Table I, there is a notable imbalance in the number of targets across different classes. In the first experimental setup, we select three classes of SAR targets from the Sentinel-1 IW mode: BulkCarrier, Cargo, and Container. These classes encompass two cross-resolution scenarios transitioning from 2.7∼3.5 m × 22 m to 20 × 22 m and from 20 × 22 m to 2.7∼3.5 × 22 m. In the second set of experimental setups, we choose three types of targets from the Gaofen-3 SL and UFS modes: BulkCarrier, Container, and Fishing. These classes involve two cross-resolution scenarios transitioning from 1 × 1 m to 1.124 × 1.728 m and from 1.124 × 1.728 m to 1 × 1 m. Finally, in the third experimental setup, we select three types of SAR targets from the Gaofen-3 UFS mode and Sentinel-1 IW mode imaging: Cargo, Fishing, and Tanker. These classes also include two cross-resolution scenarios transitioning from 1.124 × 1.728 m to 20 × 22 m and from 20 × 22 m to 1.124 × 1.728 m. The target samples under different experimental settings are illustrated in Fig. 4. It can be noticed that the differences between the same targets at different resolutions are very significant.

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TABLE II

COMPARISON OF RECOGNITION ACCURACY (%) BETWEEN OUR METHOD AND OTHER SOTA METHODS IN CROSS-RESOLUTION SCENES BASED ON THE OPENSARSHIP DATASET UNDER THE SAME IMAGING MODE CONDITIONS ON THE SAME SATELLITE

Methods	2.7~3.5m×22m→20m×22m				20×22m→2.7~3.5m×22m				Avgerage
	BulkCarrier	Cargo	Container	Avg	BulkCarrier	Cargo	Container	Avg	
CNN-Based Methods									
ResNet50 (Source Only) [15]	23.89	91.88	31.69	49.15	24.25	34.63	83.27	47.38	47.29
DAN [43]	39.66	85.59	50.70	58.65	44.28	41.52	71.97	52.59	55.62
DANN [46]	55.60	77.33	46.71	59.88	61.30	85.97	15.19	54.15	57.02
JAN [44]	15.95	78.32	75.82	56.70	52.71	77.42	36.17	55.43	56.07
ADDA [47]	16.87	84.43	73.94	58.41	77.26	75.99	11.57	54.94	56.68
CDAN [48]	31.40	88.51	58.45	59.45	47.44	67.51	44.94	53.30	56.38
MDD [49]	37.62	90.44	42.40	56.82	64.76	75.91	21.88	54.18	55.50
MCC [45]	49.88	79.62	38.73	56.08	46.69	72.05	42.86	53.87	54.98
Transformer-Based Methods									
ViT (Source Only) [32]	0.12	71.23	84.74	52.03	84.94	37.56	28.03	50.18	51.11
CDTrans [50]	56.22	79.17	44.37	59.92	44.13	58.08	72.24	58.15	59.04
SSRT [52]	56.90	80.25	53.99	63.71	65.81	62.28	45.84	57.98	60.85
TVT [51]	36.51	77.92	76.76	<u>63.73</u>	60.72	75.75	38.79	<u>58.42</u>	<u>61.08</u>
Ours	44.77	79.49	71.13	**65.13**	59.79	81.14	42.50	**61.14**	**63.14**

* Bolding indicates the best performance, underlining indicates the second best performance.

In each of our experimental setups, SAR data of a certain resolution are used as the source domain, with its class labels available, and SAR data of another resolution are used as the target domain, with its labels participating only in the test session.

All the experiments we conducted were performed on a personal computer equipped with an AMD Ryzen 9 3900X 12-core processor CPU, NVIDIA GeForce RTX 3080 GPU, and 32 GB of RAM. The code was implemented based on the PyTorch framework. Our proposed method uses LocalViT to pretrain the model on ImageNet as initialization weights and the stochastic gradient descent (SGD) algorithm as the optimizer with a batch size of 6, a momentum of 0.9, and a weight decay of 0.0001. During the training phase, the learning rate was set to 0.03, and the model was trained for 10 000 steps and evaluated every 100 steps. A warm-up phase was used for the first 500 steps of learning. In addition, the only data enhancement strategies adopted during the training phase were random flipping and center cropping. It should be noted that the experimental data were obtained by taking the average of five runs.

C. Evaluation Metric

The average accuracy, which is calculated by determining the accuracy of each class in the target domain and then averaging them, is a standard metric used to evaluate UDA classification tasks. The formula for its calculation is given as follows:

$$\text{Average Accuracy} = \frac{1}{c}\sum_{i\in c}\text{Accuracy}_i \tag{27}$$

$$\text{Accuracy}_i = \frac{\#\text{Correction}_i}{\#\text{All}_i}, \quad i = 1, 2, \ldots, c \tag{28}$$

where $\#\text{Correction}_i$ denotes the number of samples in class i that are correctly categorized and $\#\text{All}_i$ denotes the total number of samples in class i.

D. Comparisons Between the Proposed Method and Other SOTA DA Methods

This section presents comparative experimental analyses with other SOTA DA methods, in addition to a comparison with a baseline network trained solely on source domain data (source only). The methods selected for comparison include seven CNN-based methods (DAN [43], DANN [46], JAN [44], ADDA [47], CDAN [48], MDD [49], and MCC [45]) and three transformer-based methods (CDTrans [50], SSRT [52], and TVT [51]). As indicated in Tables II and III and Fig. 6, it can be initially observed that all DA methods are considerably more accurate than source only. Among these, the transformer-based methods generally exhibit slightly higher recognition performance compared to the CNN-based methods, and our method achieves the optimal results. A detailed analysis of each set of experiments will follow next.

1) Cross-Resolution Recognition Results for Three Classes of Targets From the Sentinel-1 IW Mode: Our first experiment is conducted on three elaborated classes of the Cargo class in OpenSARShip, including two cross-resolution cases, 2.7~3.5 × 22 m → 20 × 22 m and 20 × 22 m → 2.7~3.5 × 22 m. The detailed results of the experiment are presented in Table II. According to Table II, the cross-resolution recognition accuracy of case 2.7~3.5 × 22 m → 20 × 22 m is higher than case 20 × 22 m → 2.7~3.5 × 22 m in both resolutions with richer data. This is due to the fact that the knowledge that the model can learn from the high-resolution source domain, which is abundant in feature information, to the low-resolution target domain, which lacks sufficient information, is richer compared to its inverse. In addition, the

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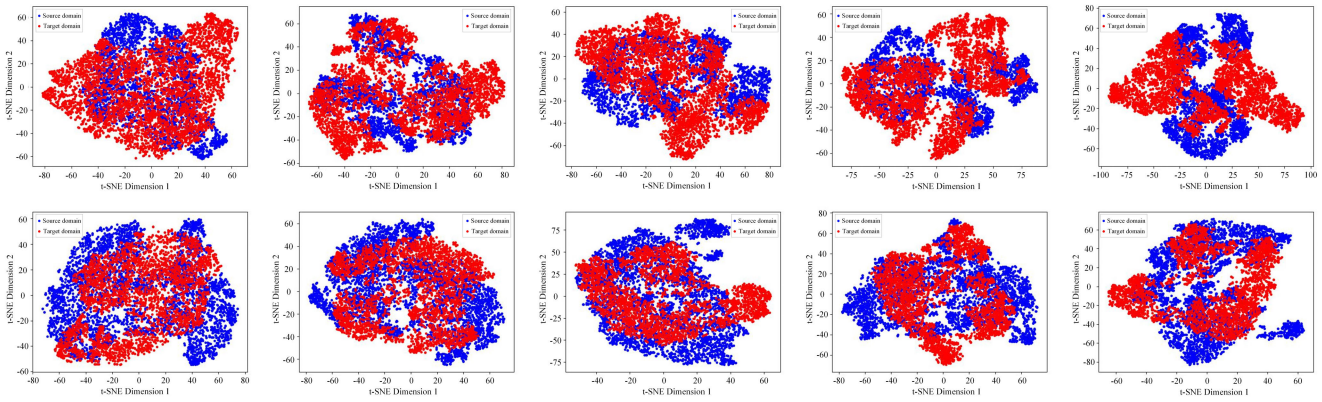


Fig. 5. t-SNE feature visualization of different recognition methods for our cross-resolution scene under the same imaging mode conditions on the same satellite. The first line of results represents 2.7∼3.5 ×22 m → 20 × 22 m, and the second line of results represents 20 × 22 m → 2.7∼3.5 × 22 m. (a) ViT (source only). (b) CDTrans. (c) SSRT. (d) TVT. (e) Ours.

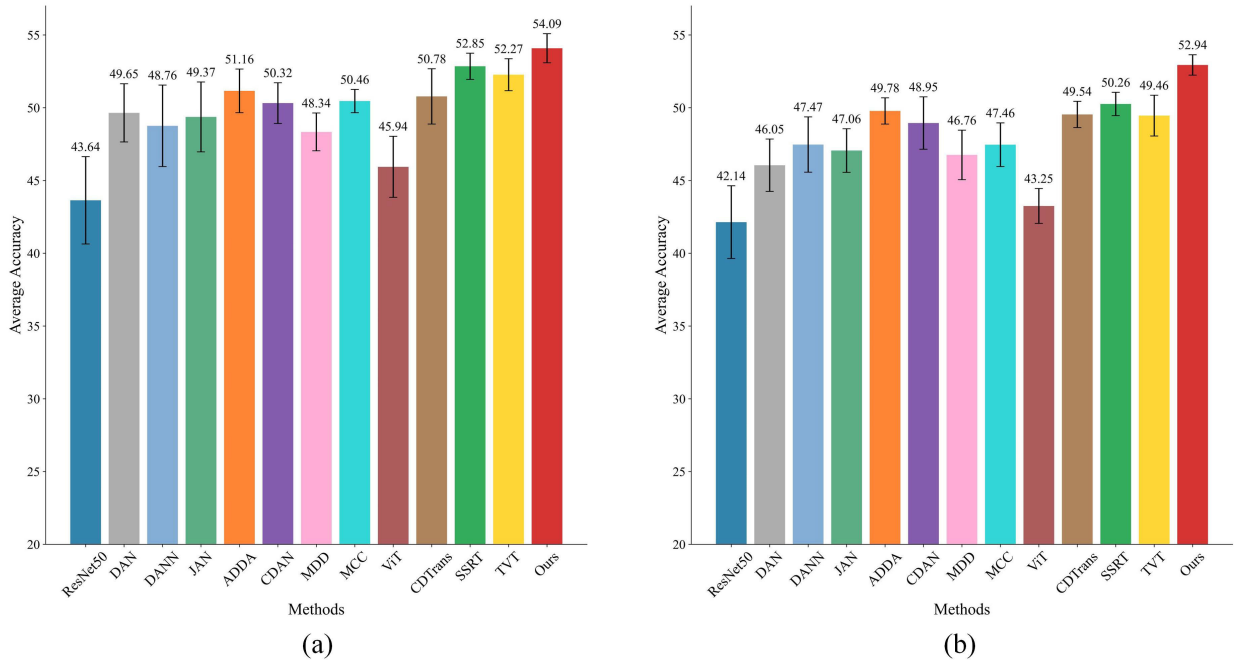


Fig. 6. Comparison accuracy (%) between our method and other methods. (a) Recognition results from 1 × 1 m resolution in the source domain to 1.124 × 1.728 m resolution in the target domain. (b) Recognition results from 1.124 × 1.728 m resolution in the source domain to 1 × 1 m resolution in the target domain.

DA methods trained with the assistance of unlabeled target domains exhibit a significant increase in accuracy across all methods compared to those trained solely on source domain data. This suggests the presence of domain transfer between data of different resolutions.

Regardless of whether it is 2.7∼3.5 × 22 m → 20 × 22 m or 20 × 22 m → 2.7∼3.5 × 22 m, transformer-based methods generally outperform CNN-based methods. This indicates that, for fine-grained recognition, the SAR target class feature differences are minimal. Thus, focusing on global semantic feature information is more advantageous for class recognition than local feature information, leading to superior results on the target domain. Moreover, our method yields the best results among these SOTA methods, with a 1.4% improvement for 2.7∼3.5 × 22 m → 20 × 22 m and a 2.72% improvement for 20 × 22 m → 2.7∼3.5 × 22 m compared to the best of these SOTA methods. This demonstrates the feasibility and effectiveness of our method in designing it from the resolution perspective of SAR.

In addition to the above quantitative analyses, we also performed t-distributed stochastic neighbor embedding (t-SNE) visualization of the target domain under two cross-resolution conditions for "ViT (source only)," "CDTrans," "SSRT," "TVT," and our method, as depicted in Fig. 5. It can be observed that the distribution of the target domain in the "source only" condition is divergent. The application of "CDTrans," "SSRT," and "TVT" can alleviate class confusion to some extent. It is noteworthy that our method is capable

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TABLE III

COMPARISON OF RECOGNITION ACCURACY (%) BETWEEN OUR METHOD AND OTHER SOTA METHODS IN CROSS-RESOLUTION SCENE BASED ON THE OPENSARSHIP DATASET AND THE FUSAR-SHIP DATASET UNDER DIFFERENT IMAGING MODE CONDITIONS FROM DIFFERENT SATELLITES

Methods	1.124m×1.728m→20m×22m				20m×22m→1.124m×1.728m				Avgerage
	Cargo	Fishing	Tanker	Avg	Cargo	Fishing	Tanker	Avg	
CNN-Based Methods									
ResNet50 (Source Only) [15]	29.40	88.72	5.43	41.18	3.03	91.86	18.93	37.94	39.56
DAN [43]	43.50	65.16	37.14	48.60	77.39	0.89	47.12	41.80	45.20
DANN [46]	65.50	30.77	52.6	49.62	82.85	11.87	33.51	42.47	46.05
JAN [44]	48.20	97.74	3.61	49.85	79.51	6.53	39.79	41.94	45.90
ADDA [47]	80.30	41.63	35.84	52.59	74.81	3.26	52.88	43.65	48.12
CDAN [48]	74.00	57.47	23.41	51.63	66.92	8.61	51.31	42.28	46.96
MDD [49]	61.10	47.96	35.40	48.15	35.96	84.87	3.66	41.50	44.83
MCC [45]	64.20	24.89	60.40	49.83	50.08	32.94	44.50	42.51	46.17
Transformer-Based Methods									
ViT (Source Only) [32]	38.30	59.28	42.77	46.78	94.08	17.80	9.95	40.61	43.70
CDTrans [50]	85.60	63.35	8.96	52.64	75.11	16.02	51.83	<u>47.65</u>	50.15
SSRT [52]	75.30	57.01	32.95	55.09	66.92	18.40	53.40	46.24	50.67
TVT [51]	69.30	93.67	4.19	<u>55.72</u>	39.45	72.40	29.32	47.06	<u>51.39</u>
Ours	86.10	76.02	8.24	**56.79**	42.64	55.70	49.21	**49.21**	**53.00**

* Bolding indicates the best performance, underlining indicates the second best performance.

of aligning class distributions more compactly, which underscores its effectiveness.

2) Cross-Resolution Recognition Results for Three Classes of Targets From the Gaofen-3 SL and UFS Modes: The second experiment includes both 1 × 1 m → 1.124 × 1.728 m and 1.124 × 1.728 m → 1 × 1 m cross-resolution cases, and the results of the comparative experiments of various methods are depicted in Fig. 6. The figure illustrates the mean and standard deviation of the results from five experiments. From the figure, it can be observed that, regardless of whether it is 1 × 1 m → 1.124 × 1.728 m or 1.124 × 1.728 m → 1 × 1 m, the recognition accuracy of ResNet50 and ViT, which only uses source domain data, is significantly lower than that of DA methods. Moreover, transformer-based DA methods generally offer greater accuracy and stability compared to CNN-based DA methods. This is largely due to CNN's excessive focus on localization, which overlooks the SAR target's overall semantic information. Despite the limited amount of data in these experiments, our method still achieves optimal results, achieving 54.09% accuracy under 1 × 1 m → 1.124 × 1.728 m condition and 52.94% under 1.124 × 1.728 m → 1 × 1 m condition. This success is attributable to the integration of overall and local information of SAR in the method design, and the improved performance of the module designed for resolution changes. Overall, the optimal accuracy of our method is improved by an average of 2.06% compared to these DA methods in both cross-resolution cases, reflecting the advancement of our method.

3) Cross-Resolution Recognition Results for Three Classes of Targets From the Sentinel-1 IW Mode and the Gaofen-3 UFS Mode: As depicted in Table III, we present the results of the cross-resolution recognition experiments conducted by our method and other SOTA methods on SAR targets under different satellites and different operating modes. Our method consistently demonstrates superior performance when the resolution across exceeds tenfold, regardless of whether CNN- or transformer-based methods are compared. When transitioning from 1.124 × 1.728 m to 20 × 22 m, our method improves by 1.07% compared to the optimal results achieved by other DA methods. Similarly, when transitioning from 20 × 22 m to 1.124 × 1.728 m, our method improves by 2.51%. These results directly underscore the effectiveness and generalizability of the method proposed in this article.

E. Ablation Studies for Our Method

In order to gain a clearer understanding and evaluate the effectiveness of the proposed method, this section will conduct a multitude of ablation experiments and analyses for each component of the proposed method. The ablation study for each module is carried out in three sets of cross-resolution experimental settings. Specifically, we compare the proposed method with the baseline method (LocalViT with global and local adversarial networks) and incrementally introduce innovative modules beyond the baseline. The detailed results are presented in Fig. 7 and Table IV.

1) Ablation Study of MSFFM: The proposed MSFFM aims to capture a multiscale semantic feature representation of SAR targets through the fusion of self-attention at different scales. This approach enhances the robust feature representation of the model for targets with varying resolutions and improves the model's adaptability to targets with varying resolutions, thereby improving performance of the target domain.

The results in Fig. 7 and Table IV reveal that within the established three sets of cross-resolution experiments, recognition performance is enhanced in both high-to-low and low-to-high resolution scenarios. However, the improvement from high to low resolution is more significant. This disparity is

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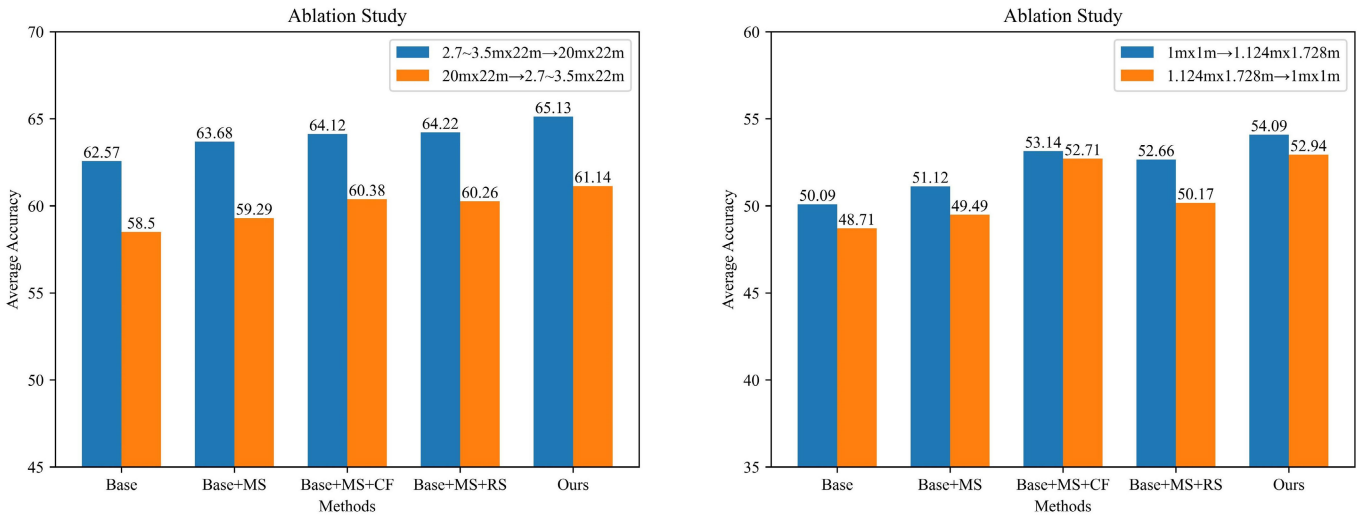


Fig. 7. Results of ablation study for each module under experimental setup 1 and experimental setup 2 conditions. (a) and (b) Results of experimental setups 1 and 2, respectively. The gradually proposed modules are MSFFM (MS), CFAM (CF), and RSCM (RS).

TABLE IV

DETAILED RESULTS OF ABLATION EXPERIMENTS FOR CROSS-RESOLUTION RECOGNITION IN EXPERIMENTAL SETUP 3 FROM DIFFERENT SATELLITES WITH DIFFERENT IMAGING MODES COMPARING THE BASELINE METHOD AND EACH MODULE

Methods	MSFFM	CFAM	RSCM	1.124m×1.728m→20m×22m				20m×22m→1.124m×1.728m				Avgerage
				Cargo	Fishing	Tanker	Avg	Cargo	Fishing	Tanker	Avg	
Baseline				66.20	85.07	5.20	52.16	65.55	68.84	1.57	45.32	48.74
Ours	✓			73.40	85.07	3.18	53.88	27.31	89.32	19.90	45.51	49.70
	✓	✓		68.20	93.67	6.94	56.27	9.56	93.18	39.79	47.51	51.89
	✓		✓	42.20	81.90	42.63	55.58	60.24	64.69	15.71	46.88	51.23
	✓	✓	✓	86.10	76.02	8.24	**56.79**	42.64	55.70	49.21	**49.21**	**53.00**

* Bolding indicates the best performance.

primarily due to the richer detail present in the high-resolution source domain, which, when processed through the MSFFM, results in target features of higher discriminative quality, thereby yielding superior recognition outcomes in the target domain. Furthermore, we also visualized the attentional weights for different classes of targets in the third set of cross-resolution experiments, as depicted in Fig. 8. From the results in Fig. 8, it can be observed that our method pays more attention to the target region features compared to the baseline, and it is more resistant to background clutter. This validation confirms the effectiveness of the proposed MSFFM in cross-resolution recognition scenarios and emphasizes its capability to extract robust features of the target.

2) Ablation Study of CFAM: The proposed CFAM aims to enhance the class differentiation among SAR targets by projecting target features of different resolutions into a common feature space within the hypersphere, followed by intraclass aggregation and interclass separation. This approach effectively improves the high interclass similarity caused by the SAR imaging mechanism and mitigates the class feature confusion caused by adversarial network feature alignment, thereby enhancing the network's recognition performance for the unlabeled target domain.

As depicted in the results in Fig. 7 and Table IV, the accuracy of the method with the inclusion of CFAM on the target domain is significantly improved under the three sets of experimental setup conditions. The results of the attention weight visualization in Fig. 8 further illustrate that the model pays attention to more discriminative target features after adding CFAM. In addition, we conducted experiments on the impact of different values of the hyperparameter ξ of CFAM on the model performance under the conditions of the third set of experimental settings, and the results are displayed in Fig. 9(a). As observed in Fig. 9(a), the best performance is achieved when the value of parameter ξ is 0.05.

3) Ablation Study of RSCM: The proposed RSRM aims to enhance the adaptability to the semantic information in the target domain by constructing multivariate Gaussian distributions for each class of different resolution SAR targets and using the sampling direction as the direction of semantic compensation for resolution. It can effectively convert the degree of attention to high- and low-frequency information of different resolution targets, thus enhancing the recognition ability in the target domain.

The results in Fig. 7 and Table IV show that there is a significant improvement in the experimental results by adding

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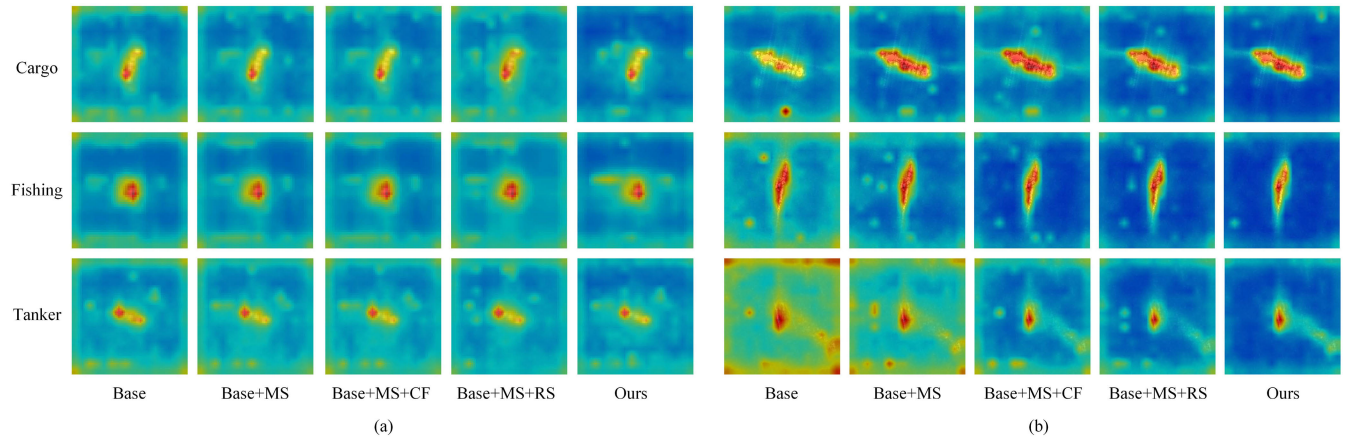


Fig. 8. Visualization of attentional weights for the ablation experiments of the innovation module and the baseline methodology under experimental setup 3. (a) Recognition results from 1.124 × 1.728 m resolution in the source domain to 20 × 22 m resolution in the target domain. (b) Recognition results from 20 × 22 m resolution in the source domain to 1.124 × 1.728 m resolution in the target domain.

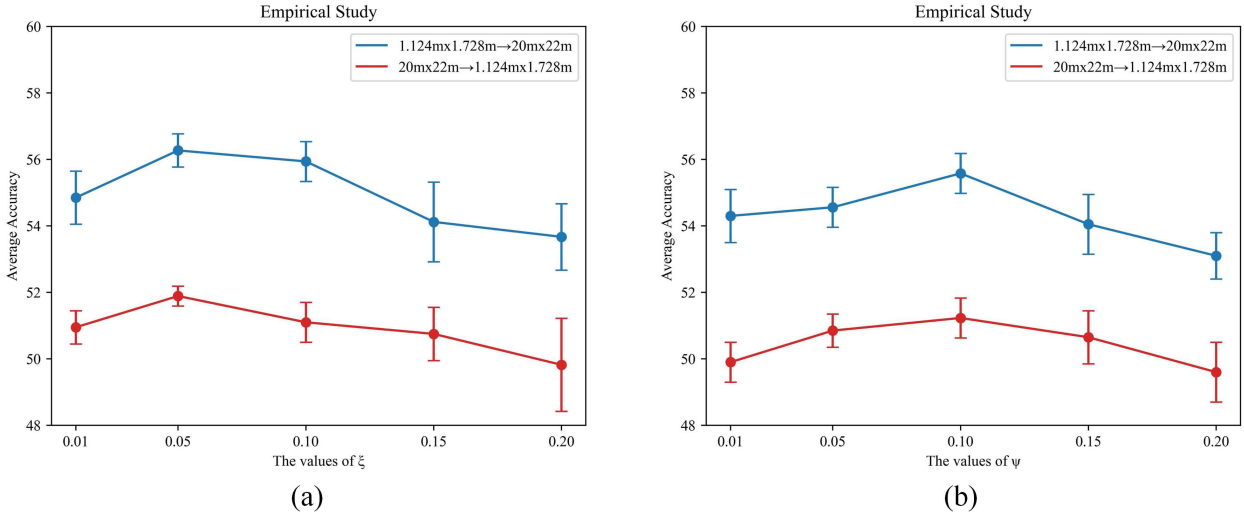


Fig. 9. Empirical study. (a) and (b) Empirical study based on the addition of MSFFM for hyperparameter ξ of CFAM and hyperparameter ψ of RSCM, respectively.

RSCM in each set of experimental setup conditions, especially in the third group of experimental setup conditions, with an average improvement of 1.63%. The results of the visualization of the attention weights in Fig. 8 reflect the fact that the model focuses more on the target after adding RSCM, which facilitates the recognition of the target. Meanwhile, we also ablated the effect of different values of the hyperparameter ψ of the RSCM on the accuracy under the third set of experimental conditions. As can be seen from Fig. 9(b), the performance is optimal when the parameter ψ takes the value of 0.10.

4) Conclusion of Ablation Study: In this subsection, we conducted ablation studies on the proposed innovation modules and analyzed each part in detail individually. As can be seen in Fig. 7 and Table IV, the effectiveness of the joint innovative modules is substantially improved. Also, the visualization of the attention weights in Fig. 8 visualizes the effectiveness of each part of our method. The visualization results show that compared with the baseline network, our method is able to effectively differentiate between the background and the target and focus more attention on the target. In addition to this, in the absence of strategies such as dealing with zero padding as mentioned in literature [70], [71], our method is equally effective in reducing edge artifacts through careful processing of features and compensation of semantic information, which further illustrates the effectiveness of the proposed method.

V. Conclusion

In this article, we introduce a DA framework based on FAC aimed at addressing the challenge of SAR image target recognition across varying resolutions. Initially, we establish global and local adversarial networks, grounded in a unique transformer-LocalViT as the baseline, to capture invariant features under cross-resolution conditions. Subsequently, we enhance the extraction of robust target features by incorporating self-attention within LocalViT through the design of an MSFFM. Next, we propose a CFAM to consolidate similar sample features and differentiate dissimilar ones, thereby boosting the discriminatory capability of the target. Finally, we introduce an RSCM to counteract the loss of semantic

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information about resolution during target recognition across varying resolutions by converting high- and low-frequency information. Experiments on three datasets, OpenSARShip, FUSAR-Ship, and SRSDD-v1.0, demonstrate that our method outperforms current SOTA DA methods. However, the method only considers cross-resolution target recognition SAR scenes from a single resolution to a single resolution, and in practical scenarios, SAR data often consist of overlapping data from multiple resolutions. Therefore, in the future, we will consider more complex cross-resolution experimental scenarios from multiresolution to multiresolution.

ACKNOWLEDGMENT

The authors would like to thank the editors and reviewers for their insightful and helpful comments.

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Zongyong Cui (Member, IEEE) received the B.E. and Ph.D. degrees in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2007 and 2015, respectively.

From 2013 to 2014, he was a Visiting Student with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore. He is currently an Associate Professor with the School of Information and Communication Engineering, UESTC. His research interests include radar image processing, target recognition, and machine learning.

Dr. Cui is a reviewer of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and *IET Radar, Sonar & Navigation*.

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Kailing Tang (Student Member, IEEE) received the B.E. degree in information engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2022, where he is currently pursuing the M.S. degree with the School of Information and Communication Engineering.

His research interests include synthetic aperture radar (SAR) image processing and object recognition.



Zheng Zhou (Graduate Student Member, IEEE) received the B.E. degree from Chengdu University of Technology, Chengdu, China, in 2019. He is currently pursuing the Ph.D. degree with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu.

His research interests include synthetic aperture radar (SAR) image processing, target detection, and few-shot learning.



Zhiyuan Yang (Student Member, IEEE) received the B.E. degree in communication engineering from Zhengzhou University, Zhengzhou, China, in 2022. He is currently pursuing the M.S. degree with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu, China.

His research interests include machine learning and synthetic aperture radar (SAR) image processing.



Liqiang Mou (Student Member, IEEE) received the B.E. degree from Chengdu University of Technology, Chengdu, China, in 2022. He is currently pursuing the M.S. degree with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu.

His research interests include synthetic aperture radar (SAR) image processing, target recognition, and SAR target multidimensional feature fusion.



Zongjie Cao (Member, IEEE) received the B.E. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1999 and 2005, respectively.

From 2006 to 2008, he was a Post-Doctoral Researcher with the Communication and Information System Postdoctoral Center, University of Electronic Science and Technology of China (UESTC), Chengdu, China. In 2008, he joined the School of Electronic Engineering, UESTC. He is currently a Professor with the School of Information and Communication Engineering, UESTC. He has authored or coauthored over 100 articles. His research interests include radar signal processing, synthetic aperture radar imaging, and target recognition.

Dr. Cao is a reviewer for several international journals and conferences, such as *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and *Remote Sensing*.



Jianyu Yang (Member, IEEE) received the B.S. degree from the National University of Defense Technology, Changsha, China, in 1984, and the M.S. and Ph.D. degrees from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1987 and 1991, respectively.

He is currently a Professor with UESTC. His research interests include synthetic aperture radar and statistical signal processing.

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